



BUSINESS ANALYSIS

Lecture 9: Business Process Analysis

1



PROCESS AND BUSINESS PROCESS ANALYSIS

- **What is a process**

“A process is a set of related activities, along with supporting information such as data and content. The activities can be part of a structured flow, or ad-hoc activities that are not part of a structured flow.”

- **What is business process analysis**

“Business process analysis (BPA) is an approach to analyzing business operation processes. It is a detailed, multi-step examination of each part of a process to identify what is working well in your current process, what needs to be improved and how any necessary improvements can best be made. There are different business process analysis methods, but all apply the underlying principle that optimized systems generate better overall business results.”

Source: IBM; <https://www.ibm.com/docs/en/dbaoc?topic=overview-what-is-process>
IBM; <https://www.ibm.com/think/topics/business-process-analysis>

2



BUSINESS PROCESS ANALYSIS

- **Methods of BPA**

Six Sigma: A five- to seven-step methodology that most businesses today use to analyze efficiencies and restraints

Lean Six Sigma: Combination of Six Sigma & Lean philosophy; focuses on eliminating tasks & resources that don't provide value

- **Steps involved in BPA**

Define: Start by identifying the processes you want to analyze.

Measure: Next, review how the process functions against defined metrics.

Analyze: There are several types of analysis techniques, and each one serves a different purpose.

Improve: Business process managers collaborate with analysts to create and execute plans that improve problem areas.

Control: After such a significant analysis, controlling the new standards and processes is the final step.

Source: IBM; <https://www.ibm.com/think/topics/business-process-analysis>

3



PROCESS AND BUSINESS PROCESS ANALYSIS

- **AS-IS process**

Overview of a company's current processes, culture, and capabilities; documents how work is currently done and how information flows throughout the organization

Purpose:

- To identify areas of improvement, optimize resource utilization for boosting efficiency and productivity
- To eliminate bottlenecks and reduce waste

- **TO-BE process**

Outlines the future state of a company's processes, culture, and capabilities; serves as a roadmap for change, defining the organization's goals and what it needs to do to achieve them.

Purpose:

- To provide a vision of the future and to identify the steps required to get there.
- To prioritize the most critical changes needed to close the gap between current and desired states

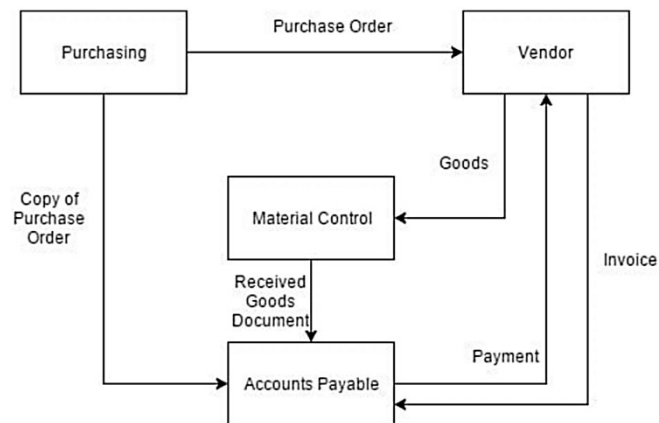
Source: Visual Paradigm; <https://www.visual-paradigm.com/tutorials/as-is-to-be-business-process.jsp>

4



BUSINESS PROCESS RE-ENGINEERING AT FORD IN THE 1980s

As-Is manual workflow in the Accounts Payable Department



Source: slideshare; <https://www.slideshare.net/slideshow/ford-motors-company-business-process-reengineering/196560893>

SIXSIGMA INSTITUTE:

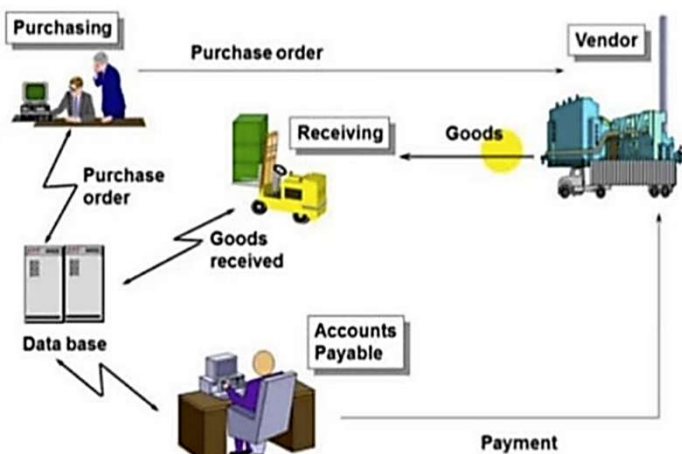
https://www.sixsigma-institute.org/Six_Sigma_Vs_Business_Process_Reengineering_Comparison.php#:~:text=One%20example%20of%20business%20process,the%20dependency%20on%20physical%20paperwork.

5



BUSINESS PROCESS RE-ENGINEERING AT FORD IN THE 1980s

To-Be Digitalized workflow in the Accounts Payable Department



Source: slideshare; <https://www.slideshare.net/slideshow/ford-motors-company-business-process-reengineering/196560893>

SIXSIGMA INSTITUTE:

https://www.sixsigma-institute.org/Six_Sigma_Vs_Business_Process_Reengineering_Comparison.php#:~:text=One%20example%20of%20business%20process,the%20dependency%20on%20physical%20paperwork.

6



BUSINESS PROCESS RE-ENGINEERING AT FORD IN THE 1980s

The result of BPR at the Accounts Payable Department at Ford

- **Cost:** Reduced workforce by 75%, hence salary expenses
- **Speed:** Documents processing time for the Accounts Payable Department was reduced, which also had a positive impact on relationship with vendors/suppliers
- **Quality:** Digitalization reduced manual errors
- **Service:** The company's productivity increased as there are no more delays in the purchasing of the raw materials.

Source: studocu; <https://www.studocu.com/row/document/riphah-international-university/international-business/bpr-ford-motors-i-love-studocu-website/31670260>

7



TOOLS FOR PROCESS ANALYSIS: FLOWCHARTS

• Definition

"A flowchart is a diagram that depicts the stages of a process, workflow, computer program or system. Flowchart diagrams consist of shapes connected by lines and represent step-by-step processes to aid in decision-making, reduce ambiguity and enhance workflows."

• What are flowcharts used for

- "Documenting a process or procedure
- Brainstorming ideas
- Business process management (BPM) and business process analysis (BPA)
- Increasing clarity and improving communication between colleagues and teams
- Outlining a decision-making procedure
- Clarifying the reporting structure of an organization
- Optimizing a process or workflow"

Source: IBM; <https://www.ibm.com/think/topics/flowchart>

8



TYPES OF FLOWCHARTS

- **Program flowcharts** are the blueprints for software and computer programs
- **System flowcharts** govern the fundamental sequence of events in a system
- **Workflow diagrams** detail the process for initiating and completing a business process
- **Decision flowcharts** (decision trees) illustrate the stages and outcomes of a decision-making process
- **Data flow diagrams** (DFD) chart the flow of information through a process, network or system
- **Process flow diagrams** (PFD) illustrate the operations of chemical and process engineering facilities
- **Swimlane diagrams** indicate the personnel responsible for each stage of a workflow
- **Organizational charts** display the internal structure of a team, department or organization

Source: IBM; <https://www.ibm.com/think/topics/flowchart>

9



FLOWCHART SYMBOLS

Symbol	Name	Function
	Start/end	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectangle represents a process
	Decision	A diamond indicates a decision

Source: smartdraw; <https://www.smartdraw.com/flowchart/flowchart-symbols.htm?srsltid=AfmBOoqsYG7DWIOBkn-ofjsfjg8qbRF7RMJNSDzPIMQpbWj959Ds4ZA>

10

TOOLS FOR PROCESS ANALYSIS/MAPPING



- **Business Process Model Notation (BPMN) diagramming**

Business process modeling and notation (BPMN) is the global standard for modeling business processes. It is a fundamental part of business process management (BPM). BPMN diagrams allow stakeholders to visualize business processes, making it easier to streamline workflows.

- **How BPMN works**

The BPMN language is based on flowcharts and graphical notations. The standard notation used to represent BPMN elements is separated into several categories for diagramming

- **Flow objects:** Flow objects are descriptive objects that define a process, such as events, activities and gateways
- **Events:** Events are triggers that start, alter or conclude a process.
- **Activities:** Activities are actions or tasks performed during the business process.
- **Gateways:** Gateways are decision points requiring an active choice to be made before progressing further through the business process.
- **Connecting objects:** Connecting objects are shown as lines with arrows and are used to connect flow objects.
- **Swimlanes:** Swimlanes are containers that separate one set of activities from others.
- **Artifacts:** Artifacts provide supplemental information about the process, such as data objects, groups and annotations
- **Video: Business Process Model and Notation (BPMN) Diagrams;** <https://www.youtube.com/watch?v=bnU7eMhiMHM>

Source: IBM; <https://www.ibm.com/think/topics/bpmn>
Signavio; <https://www.signavio.com/wiki/process-design/bpmn/>

11

TOOLS FOR PROCESS ANALYSIS/MAPPING



- **Supplier, Input, Process, Output, Customer (SIPOC) model diagramming**

A SIPOC diagram is a high-level visual map that summarizes the critical parts of a process on a single page. The name SIPOC comes from the first letters of its five building blocks, Suppliers, Inputs, Process, Outputs, and Customers, which together show how work flows from the very beginning to the final recipient.

Suppliers provide the resources or information.

Inputs are the materials or data entering the process.

Process represents the sequence of activities that transform those inputs.

Outputs are the products, services, or results produced.

Customers are the people or entities receiving those outputs.

Video: SIPOC DIAGRAM; <https://www.youtube.com/watch?v=6AeJk61tIGE>

Source: Invensis; <https://www.invensislearning.com/blog/what-is-sipoc-diagram/#:~:text=A%20SIPOC%20diagram%20is%20a,or%20entities%20receiving%20those%20outputs.>

12



ACTIVITY: FLOW CHARTS

You have just had your mid-term examination, conducting which is an essential job of the University

- Download Draw.io
- From the **perspective of the University**, draw a flowchart of the AS-IS process starting from (i) a student having been given the exam timetable to (ii) the date he enters the examination hall, (iii) sits the exam, and (iv) hands over his paper and leaves the hall
- Identify the key elements of this process
- How can the University **improve the efficiency of the process for the students**? Draw a flowchart of the TO-BE process after identifying issues in the current process

Note: Remember that for the above process to be efficient

- (1) Any waste of time should be eliminated
- (2) The students should be made comfortable so that they can focus only on the job at hand i.e., solving the paper
- (3) There should be no disruptions, including any unnecessary communication between the invigilators and the students

13



ESSENTIAL READINGS

IBM: "What is a process", available at <https://www.ibm.com/docs/en/dbaoc?topic=overview-what-is-process>

IBM: "What is a flowchart", available at <https://www.ibm.com/think/topics/flowchart>

IBM: "What is business process analysis", available at <https://www.ibm.com/think/topics/business-process-analysis>

IBM: "What is business process modeling and notation (BPMN)?", available at <https://www.ibm.com/think/topics/bpmn/>

Invensis: "What Is a SIPOC Diagram?", available at <https://www.invensislearning.com/blog/what-is-sipoc-diagram/#:~:text=A%20SIPOC%20diagram%20is%20a,or%20entities%20receiving%20those%20outputs.>

Signavio: "Business Process Model and Notation (BPMN)", available at <https://www.signavio.com/wiki/process-design/bpmn/>

SIXSIGMA INSTITUTE: "Business Process Reengineering with Examples: Successful Case Studies"; available at https://www.sixsigma-institute.org/Six_Sigma_Vs_Business_Process_Reengineering_Comparison.php#:~:text=One%20example%20of%20business%20process,the%20dependency%20on%20physical%20paperwork.

slideshare: "Ford Motors Company: Business Process Reengineering"; available at <https://www.slideshare.net/slideshow/ford-motors-company-business-process-reengineering/196560893>

studocu: "BPR Case Study of Ford 04 - Lecture Notes on Change Management"; available at <https://www.studocu.com/row/document/riphah-international-university/international-business/bpr-ford-motors-i-love-studocu-website/31670260>

smartdraw: "Flowchart Symbols"; available at: <https://www.smartdraw.com/flowchart/flowchart-symbols.htm?srltid=AfmB0oqsYG7DWiOBkn-ofjsflg8qbRF7RMJNSDzPIMQpbWj959DsS4ZA>

Visual Paradigm: "How to Develop As-Is and To-Be Business Process"; available at: <https://www.visual-paradigm.com/tutorials/as-is-to-be-business-process.jsp>

14



REFERENCE BOOKS OPTIONS

- **BUSINESS ANALYSIS BODY OF KNOWLEDGE (BABOK)**; IIBA Bookstore: <https://www.iiba.org/career-resources/a-business-analysis-professionals-foundation-for-success/babok/>
- **BUSINESS ANALYSIS (Fourth Edition)** by Debra Paul (Author), James Cadle (Author), Malcolm Eva (Author), Craig Rollason (Author), Jonathan Hunsley (Author)
- **BUSINESS ANALYSIS TECHNIQUES: 123 ESSENTIAL TOOLS FOR SUCCESS (3RD Edition)** by James Cadle (Author), Debra Paul (Author), Jonathan Hunsley (Author), Adrian Reed (Author)